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True/False

1. Software engineering is a field that encompasses designing, writing, testing, debugging, documenting, modifying, and maintaining computer programs.

True

2. Pseudocode is a form of a program statement that will always evaluate to "false."

False

3. In programming, the terms "line" and "statement" always mean the same thing.

False

4. In C++, key words are written in all lowercase letters.

True

5. The preprocessor executes after the compiler.

False

6. Machine language is an example of a high-level language.

False

7. A CPU only understands machine language instructions.

True

8. Programs are often referred to as hardware.

False

9. The CPU is the most important component in a computer because without it, the computer could not run software.

True

10. The term "bit" stands for binary digit.

True

MULTIPLE CHOICE

1. What does the term *hardware* refer to?

- a. The relative difficulty of writing computer programs
- b. The physical components that make up a computer
- c. The way a computer's storage space is organized
- d. The logical flow of instructions
- e. None of these

Answer: B

2. A(n) _____ is a set of instructions that the computer follows to solve a problem.

- a. compiler
- b. linker
- c. program
- d. operator
- e. None of these

Answer: C

3. Computer programs are also known as

- a. hardware
- b. firmware
- c. software
- d. Any of these
- e. None of these

Answer: C

4. At the heart of a computer is its central processing unit. The CPU's job is:

- a. To fetch instructions
- b. To carry out the operations commanded by the instructions
- c. To produce some outcome or resultant information
- d. All of these
- e. None of these

Answer: D

5. A computer stores a program while it is running

- a. in main memory
- b. on a hard disk
- c. on the computer monitor
- d. in the CPU
- e. None of these

Answer: A

6. The _____ decodes an instruction and generates an electronic signal.

- a. Arithmetic and Logic Unit
- b. Main memory
- c. BIOS
- d. Control Unit
- e. None of these

Answer: D

7. The CPU's control unit retrieves the next instruction in a sequence of program instructions from main memory in the _____ stage.

- a. fetch
- b. decode
- c. execute
- d. portability

Answer: A

8. During which stage does the central processing unit analyze the instruction and encode it in the form of a number, and then generate an electronic signal?

- a. fetch
- b. decode
- c. execute
- d. portability

Answer: B

9. The two parts of the CPU are

- a. the output device and the input device
- b. the software and the hardware

- c. the Control Unit and the Arithmetic and Logic Unit
- d. the single-task device and the multi-task device
- e. None of these

Answer: C

10. A volatile type of memory that is used for temporary storage is

- a. an address
- b. the ALU
- c. RAM
- d. a disk drive
- e. None of these

Answer: C

11. The purpose of a memory address is:

- a. to identify the location of a byte in memory
- b. to prevent multitasking
- c. to obtain an algorithm
- d. to improve the speed of processing

- e. None of these

Answer: A

12. Programs are normally stored in _____ and loaded into main memory as needed.

- a. the input device
- b. the output device
- c. secondary storage
- d. the CPU
- e. None of these

Answer: C

13. A computer monitor is a type of

- a. input device
- b. output device
- c. storage device
- d. software
- e. None of these

Answer: B

14. Which of the following is *not* a common input device?

- a. keyboard
- b. mouse
- c. digital camera
- d. printer
- e. All are common input devices

Answer: D

15. Which of the following is *not* one of the major components of a computer system?

- a. the preprocessor
- b. the CPU
- c. main memory
- d. input/output devices
- e. secondary storage

Answer: A

16. A set of well-defined steps for performing a task or solving a problem is known as a(n):

- a. hierarchy chart
- b. algorithm
- c. instruction set
- d. statement
- e. None of these

Answer: B

17. When a programmer saves to a file the statements he or she writes to create a program, these statements are

- a. high level
- b. source code
- c. a preprocessor file
- d. object code
- e. None of these

Answer: B

18. The programmer usually enters source code into a computer with

- a. a hierarchy chart

- b. a text editor
- c. a compiler
- d. pseudocode
- e. None of these

Answer: B

19. In the process of translating a source file into an executable file, which of the following is the correct sequence?

- a. Source code, preprocessor, modified source code, linker, object code, compiler, executable code
- b. Preprocessor, source code, compiler, executable code, linker, modified source code, object code
- c. Source code, compiler, modified source code, preprocessor, object code, linker, executable code.
- d. Source code, preprocessor, modified source code, compiler, object code, linker, executable code.
- e. Source code, linker, object code, compiler, modified source code, preprocessor, executable code.

Answer: D

20. An Integrated Development Environment (IDE) typically consists of

- a. a text editor
- b. a compiler
- c. a debugger
- d. All of the above
- e. None of these

Answer: D

21. _____ are used to translate each source code instruction into the appropriate machine language instruction.

- a. modules
- b. runtime libraries
- c. compilers
- d. preprocessor directives
- e. None of these

Answer: C

22. This is a set of rules that must be followed when constructing a program:

- a. syntax

- b. punctuation
- c. key words
- d. operators
- e. identifiers

Answer: A

23. Words that have a special meaning and may be used only for their intended purpose are known as

- a. operators
- b. programmer defined words
- c. key words
- d. syntax
- e. none of these

Answer: C

24. Which of the following best describes an operator?

- a. An operator is a rule that must be followed when constructing a program.
- b. An operator allows you to perform operations on one or more pieces of data.

- c. An operator marks the beginning or ending of a statement, or is used to separate items in a list.
- d. An operator is a word that has a special meaning.
- e. An operator is a symbolic name that refers to a variable.

Answer: B

25. This is used in a program to mark the beginning or ending of a statement, or separate items in a list:

- a. separators
- b. punctuation
- c. operators
- d. key words
- e. None of these

Answer: B

26. Characters or symbols that perform operations on one or more operands are:

- a. separators
- b. op codes

- c. operators
- d. key words
- e. None of these

Answer: C

27. This is a complete instruction that causes the computer to perform some action:

- a. line
- b. statement
- c. variable
- d. key word
- e. None of these

Answer: B

28. A named storage location in the computer's memory that holds a piece of information is a(n):

- a. variable
- b. operator
- c. key word

- d. statement
- e. None of these

Answer: A

29. A variable definition defines the name of a variable that will be used in a program, as well as:

- a. the type of data it will be used to hold
- b. the operators that will be used on it
- c. the number of times it will be used in the program
- d. the value it will hold
- e. None of these

Answer: A

30. Three primary activities of a program are:

- a. variable definitions, operators, lists of key words
- b. lines, statements, punctuation
- c. input, processing, output
- d. integer, floating-point, character definitions

- e. None of these

Answer: C

31. Which step uncovers any syntax errors in your program?

- a. editing
- b. compiling
- c. linking
- d. executing
- e. None of these

Answer: B

32. Mistakes that cause a running program to produce incorrect results are called:

- a. syntax errors
- b. logic errors
- c. compiler errors
- d. linker errors
- e. None of these

Answer: B

33. The programming process consists of several steps, which include:

- a. key words, operator definitions, punctuation
- b. design, creation, testing, debugging
- c. input, processing, output
- d. syntax, logic, error handling
- e. None of these

Answer: B

34. The first step in writing a program is to

- a. type the code
- b. visualize the program running on a computer
- c. visualize logical errors
- d. clearly define what the program is to do
- e. None of these

Answer: D

35. A model often used when creating a program that begins with the overall task and refines it into smaller subtasks is a(n)

- a. flowchart
- b. UML diagram
- c. blueprint
- d. hierarchy chart
- e. None of these

Answer: D

36. The term that refers to the programmer reading the program from the beginning and stepping through each statement is

- a. pseudocoding
- b. software engineering
- c. desk checking
- d. spot checking
- e. None of these

Answer: C

37. The two methods used by C++ to write computer programs are:

- a. top-down programming and procedural programming
- b. procedural programming and object-oriented programming
- c. pseudocoding and object-oriented programming
- d. flowcharting and procedural programming
- e. None of these

Answer: B